

This practical can be performed in any of the compiler like Turbo C/ vi editor etc or Python Interpreter / IDE like any open source IDE such IDLE, Jupyter Notebook, Spyder, PyCharm etc.)

(X) Conclusion

Successfully implemented C/Python program on First In First Out (FIFO) Page Replacement algorithm.

(XI) Practical Related Questions

Note: Below given are few sample questions for reference. Teachers must design more such questions to ensure the achievement of identified CO.

- (1) What is page hit ?
- (2) What is page fault?
- (3) What is FIFO page replacement? How does it work?
- (4) State the advantages and disadvantages of FIFO.
- (5) Count number of page hits in above mention example given in section IX.
- (6) Given the following page reference string: 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 How many page faults will occur using FIFO with 3 frames?

[Space to Write for Answers]

- 1) A page hit occurs when the page requested by the CPU is already present in the main memory (RAM). Since the required page is available, it can be accessed immediately without loading it from secondary storage (disk).
- 2) A page fault occurs when the page requested by the CPU is not present in the main memory (RAM). The operating system must fetch the required page from secondary storage (disk) into RAM before the process can continue execution.
- 3) FIFO Page Replacement is a page replacement algorithm in which the page that has been in memory the longest (the first page loaded) is replaced when a new page needs to be loaded and all memory frames are full.

Working :-

1. Pages are loaded into memory in the order they arrive.

2. When a page fault occurs and all frames are full :
- The oldest page (the one loaded first) is removed.
 - The new page is loaded into the freed frame.

3. This process continues using the First In First Out (FIFO) rule.

4) Advantages :-

1. Simple and easy to implement
2. Low overhead
3. Fairly replaces the oldest page first

Disadvantages :-

1. May replace frequently used pages.
2. Does not consider page usage or recency.
3. Can result in higher number of page faults than other algorithms.

5) Given :-

Reference String :- 0, 1, 2, 3, 0, 1, 2, 3, 0, 1, 2, 3, 4, 5, 6, 7

Frame size :- 3

Page Replacement Algorithm :- FIFO

RS	0	1	2	3	0	1	2	3	0	1	2	3	4	5	6	7
F ₁	0	0	0	3	3	3	2	2	2	1	1	1	4	4	4	7
F ₂		1	1	1	0	0	0	3	3	3	2	2	2	5	5	5
F ₃			2	2	2	1	1	1	0	0	0	3	3	3	6	6
H/F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F	F

Page Hits = 0

Page Faults = 16

6) Given :-

Reference String :- 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5

Frame Size :- 3

Page Replacement Algorithm :- FIFO

RS	1	2	3	4	1	2	5	1	2	3	4	5
F ₁	1	1	1	4	4	4	5	5	5	5	5	5
F ₂		2	2	2	1	1	1	1	1	3	3	3
F ₃			3	3	3	2	2	2	2	2	4	4
H/F	F	F	F	F	F	F	F	H	H	F	F	H

Page Hits = 3

Page Faults = 9